AI Use Disclosure

QryptoNox: Quantum Oracle Forensics

Team Name: QryptoNox Team

1. Use of Artificial Intelligence

Generative AI tools were used as supporting aids during the development and documentation of this project. Their use was intended to assist with understanding technical concepts, exploring implementation approaches, debugging, and preparing the written documentation. AI-generated suggestions were treated as assistance and were not considered a substitute for testing or experimental verification.

2. Areas of AI Assistance

AI assistance was used in the following areas:

- **Conceptual understanding:** Clarifying quantum computing concepts and the principles behind quantum-oracle algorithms and oracle-family identification.
- **Implementation support:** Discussing possible algorithmic approaches, reviewing code logic, and reasoning about potential errors and edge cases.
- **Debugging:** Interpreting error messages and considering possible corrections during development.
- **Documentation:** Organizing the technical report, describing the experimental setup, summarizing observed results, and improving the clarity of the written material.

3. Human Contribution and Verification

The project was implemented and evaluated in the challenge environment. The solver was tested using the provided evaluator, and the numerical results included in the technical report were taken from the observed baseline evaluation output.

AI assistance did not replace execution of the evaluator or verification of the reported numerical results. The team is responsible for the submitted implementation, its behaviour, the interpretation of the experimental findings, and compliance with the challenge requirements.

4. Limitations

AI-generated suggestions can be incorrect, incomplete, or unsuitable for specialized technical problems. Accordingly, proposed approaches and explanations were considered in the context of the actual implementation and available evaluation results.

The reported findings are preliminary and limited to experiments actually completed. Query-budget sweeps, additional noise-stress tests, and baseline comparisons that were not completed are not presented as completed experiments.

5. Declaration

We disclose the use of generative AI for conceptual assistance, implementation-related discussion, debugging support, and technical documentation. The team accepts responsibility for the final submission and for ensuring that AI use complies with the competition rules.

Team: QryptoNox Team
Date: 16 October 2026